

# Lecture 14: Independent Variables

Example: Gases,  $f(P, V, T)$  where  $PV = nRT$  or  $f(x, y, z)$  where  $g(x, y, z) = c$  If  $g(x, y, z) = c$  then  $z = z(x, y)$   $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$ ? How do we find  $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$  without solving?

Ex:  $x^2 + yz + z^3 = 8$  at  $(2, 3, 1)$  Take differential:  $2x dx + z dy + (y + 3z^2) dz = 0$  at  $(2, 3, 1)$ :  $4 dx + dy + 6 dz = 0$  If we want to view  $z = z(x, y)$ :  $dz = \frac{1}{6}(4 dx + dy)$

$$\frac{dz}{dx} = \frac{-4}{6} = -\frac{2}{3} \quad y \text{ constant, set } dy = 0$$

$$\frac{dz}{dy} = \frac{-1}{6}; x \text{ constant, set } dx = 0$$

In general:  $g(x, y, z) = c$ , then  $dg = g_x dx + g_y dy + g_z dz = 0$

$$\text{Solve for } dz = \frac{-g_x}{g_z} dx - \frac{g_y}{g_z} dy$$

So, for  $\frac{dz}{dx}$ : set  $y = \text{constant} \Rightarrow dy = 0$

$$dz = \frac{-g_x}{g_z} dx, \quad \frac{dz}{dx} = \frac{-g_x}{g_z}$$

Say  $f(x, y) = x + y$ .  $\frac{\partial f}{\partial z} = 1$

set  $x = u$ ; then  $f = x + y = 2u + v$

$$\frac{\partial f}{\partial u} = 2$$

So  $x = u$  but  $\left(\frac{\partial f}{\partial x} \neq \frac{\partial f}{\partial u}\right) \Rightarrow$  change  $u = x$  Keep  $v = y - x$  constant

We need clearer notation:

$$\left(\frac{\partial f}{\partial z}\right)_y = \text{keep } y \text{ constant}$$

$$\left(\frac{\partial f}{\partial z}\right)_y \neq \left(\frac{\partial f}{\partial z}\right)_x = \left(\frac{\partial f}{\partial y}\right)_x$$

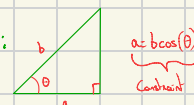
$$\left(\frac{\partial f}{\partial y}\right)_x = \text{keep } x \text{ constant}$$

Example: Area of triangle



$$A = \frac{1}{2} ab \sin \theta$$

Assume it's a right triangle:



Rate of change of  $A$  with respect to  $\theta$ ?

1) Treat  $a, b, \theta$  as independent:  $\frac{\partial A}{\partial \theta} = \left(\frac{\partial A}{\partial \theta}\right)_{a,b}$  Keep  $a, b$  fixed  $\frac{\partial A}{\partial \theta} = \frac{1}{2} ab \cos(\theta)$

2) Keep "a" constant, "b" will change:  $b = b(a, \theta) = \frac{a}{\cos \theta}$  so that we keep a right angle:  $\left(\frac{\partial A}{\partial \theta}\right)_a$

3) Keep "b" constant,  $a = a(b, \theta)$  changes  $\left(\frac{\partial A}{\partial \theta}\right)_b$  Compute  $\left(\frac{\partial A}{\partial \theta}\right)_a$ ?

Method 1: Solve for  $a$  and  $b$  and substitute

$$a = b \cos \theta \Rightarrow b = \frac{a}{\cos \theta} = \frac{a \sec \theta}{1} \quad A = \frac{1}{2} ab \sin \theta = \frac{1}{2} \frac{a^2 \sin \theta}{\cos \theta} = \frac{1}{2} a^2 \tan \theta$$

$$\text{So } \left(\frac{\partial A}{\partial \theta}\right)_a = \frac{1}{2} a^2 \sec^2 \theta \quad \frac{\partial}{\partial \theta} \tan = \sec^2 = 1 + \tan^2$$

2 systematic methods:

① Differentials

\* keep "a" fixed:  $da = 0$

\* constraint  $a = b \cos \theta \Rightarrow da = \cos \theta db - b \sin \theta d\theta$

$$0 = da = \cos \theta db - b \sin \theta d\theta \Rightarrow \cos \theta db = b \sin \theta d\theta \Leftrightarrow db = b \tan \theta d\theta$$

\* function  $A = \frac{1}{2} ab \sin \theta \Rightarrow dA = \frac{1}{2} b \sin \theta da + \frac{1}{2} a \sin \theta db + \frac{1}{2} ab \cos \theta d\theta$

$$\text{So } dA = \frac{1}{2} a \sin \theta \underbrace{(b \tan \theta d\theta)}_{db} + \frac{1}{2} ab \cos \theta d\theta = \frac{1}{2} ab \underbrace{(\sin \theta \tan \theta + \cos \theta)}_{\sec \theta} d\theta$$

$$\text{So } \left(\frac{\partial A}{\partial \theta}\right)_a = \frac{1}{2} ab \sec \theta$$

② Chain rule:  $\left(\frac{\partial}{\partial \theta}\right)_a$  in formula for  $A$ .

$$\left(\frac{\partial A}{\partial \theta}\right)_a = A \cdot \underbrace{\left(\frac{\partial \theta}{\partial \theta}\right)_a}_{=1} + A \cdot \underbrace{\left(\frac{\partial a}{\partial \theta}\right)_a}_{=0} + A \cdot \left(\frac{\partial b}{\partial \theta}\right)_a \rightarrow \text{use constraint}$$